

LAYER: InstructPix2Pix Identity-Layer Geometry Results

Stage-B targeted layer optimization with Stage-A identity-layer context

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Objective: $Z = 1 - \text{cosine_similarity}(\text{pool}(\text{layer}(\text{original})), \text{pool}(\text{layer}(\text{perturbed})))$. Loss: $\text{loss} = -Z$. No visual counter-loss was used.

Run matrix

layers	prompts	images	runs	iterations
4.0000	2.0000	4.0000	32.000	400

Aggregate results

layer	prompt	runs	mean dZ	max dZ	mean SSIM	min SSIM	invalid
unet.conv_in	add black sunglasses	4.0000	0.2115	0.3292	0.302	0.04973	3.0000
unet.conv_in	add headphones	4.0000	0.2127	0.3291	0.2742	0.01206	3.0000
unet.mid_block	add black sunglasses	4.0000	0	0	0.9802	0.9768	0
unet.mid_block	add headphones	4.0000	0	0	0.9802	0.9768	0
unet.up_blocks.0	add black sunglasses	4.0000	0	0	0.9802	0.9768	0
unet.up_blocks.0	add headphones	4.0000	0	0	0.9802	0.9768	0
vae_image_latent	add black sunglasses	4.0000	0.0001023	0.0003517	0.971	0.9547	0
vae_image_latent	add headphones	4.0000	0.002834	0.01128	0.9333	0.8164	1.0000

Per-run final values

validity	layer	prompt	image	dZ	SSIM	out SSIM	max disp
invalid_input_destroyed	unet.conv_in	add black sunglasses	barack_obama_01	0.3292	0.04973	-0.004205	8.0000
invalid_input_destroyed	unet.conv_in	add headphones	barack_obama_01	0.3291	0.0161	-0.05994	8.0000
invalid_input_destroyed	unet.conv_in	add headphones	kamala_harris_01	0.3019	0.01206	-0.02564	8.0000
invalid_input_destroyed	unet.conv_in	add black sunglasses	kamala_harris_01	0.3007	0.08558	-0.0164	8.0000
invalid_input_destroyed	unet.conv_in	add headphones	joe_biden_01	0.2196	0.09114	0.1068	8.0000
invalid_input_destroyed	unet.conv_in	add black sunglasses	joe_biden_01	0.2163	0.09515	-0.3214	8.0000
invalid_fft_saturated	vae_image_latent	add headphones	joe_biden_01	0.01128	0.8164	0.8005	7.9569
valid_or_visually_check	vae_image_latent	add black sunglasses	joe_biden_01	0.0003517	0.9674	0.9733	3.9636
valid_or_visually_check	vae_image_latent	add black sunglasses	barack_obama_01	5.74e-05	0.9547	0.9485	2.3085
valid_or_visually_check	vae_image_latent	add headphones	barack_obama_01	5.71e-05	0.9547	0.9066	2.3085
valid_or_visually_check	unet.conv_in	add black sunglasses	michelle_obama_01	0	0.9776	0.9809	1.9212
valid_or_visually_check	unet.conv_in	add headphones	michelle_obama_01	0	0.9776	0.9681	1.9212
valid_or_visually_check	unet.mid_block	add black sunglasses	barack_obama_01	0	0.9768	0.9749	1.9212
valid_or_visually_check	unet.mid_block	add black sunglasses	joe_biden_01	0	0.9819	0.9781	1.9212
valid_or_visually_check	unet.mid_block	add black sunglasses	kamala_harris_01	0	0.9844	0.9859	1.9212
valid_or_visually_check	unet.mid_block	add black sunglasses	michelle_obama_01	0	0.9776	0.9809	1.9212
valid_or_visually_check	unet.mid_block	add headphones	barack_obama_01	0	0.9768	0.9311	1.9212
valid_or_visually_check	unet.mid_block	add headphones	joe_biden_01	0	0.9819	0.9622	1.9212
valid_or_visually_check	unet.mid_block	add headphones	kamala_harris_01	0	0.9844	0.9813	1.9212
valid_or_visually_check	unet.mid_block	add headphones	michelle_obama_01	0	0.9776	0.9681	1.9212
valid_or_visually_check	unet.up_blocks.0	add black sunglasses	barack_obama_01	0	0.9768	0.9749	1.9212
valid_or_visually_check	unet.up_blocks.0	add black sunglasses	joe_biden_01	0	0.9819	0.9781	1.9212
valid_or_visually_check	unet.up_blocks.0	add black sunglasses	kamala_harris_01	0	0.9844	0.9859	1.9212
valid_or_visually_check	unet.up_blocks.0	add black sunglasses	michelle_obama_01	0	0.9776	0.9809	1.9212

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Image strips

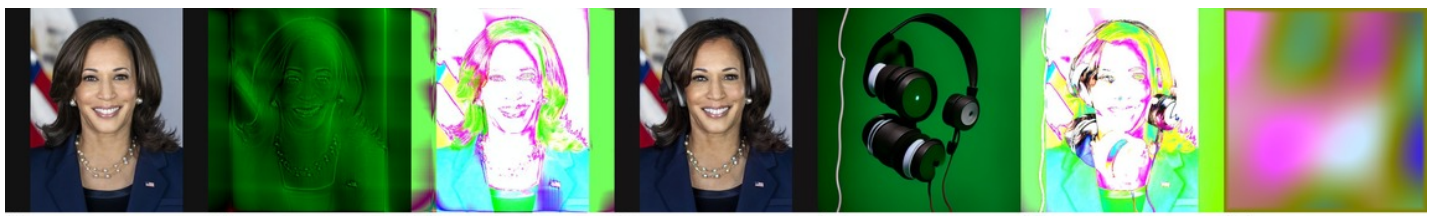
UNET.conv_in / add black sunglasses / barack_obama_01 - invalid_input_destroyed



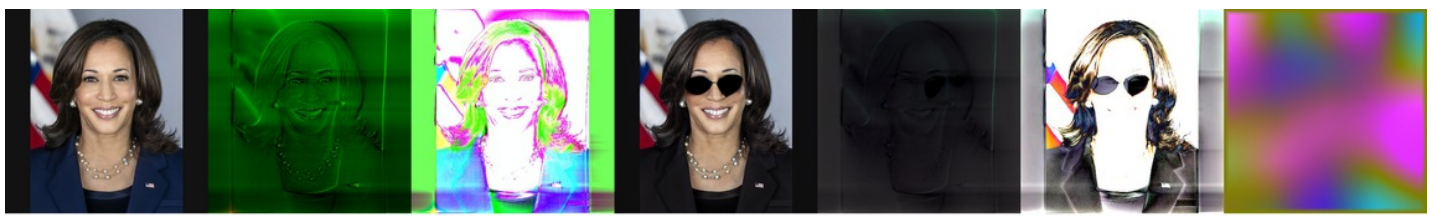
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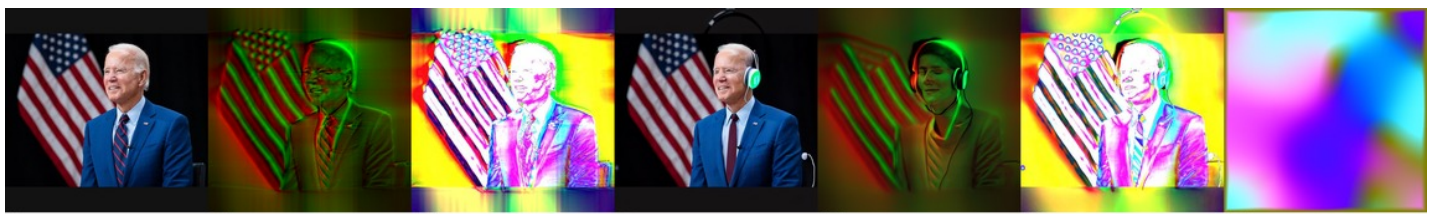
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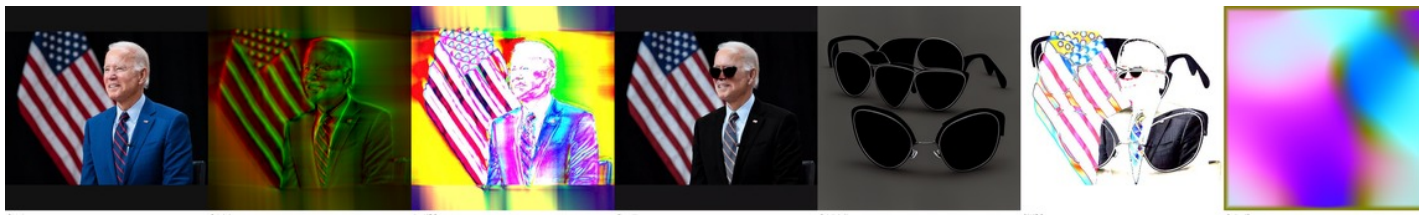
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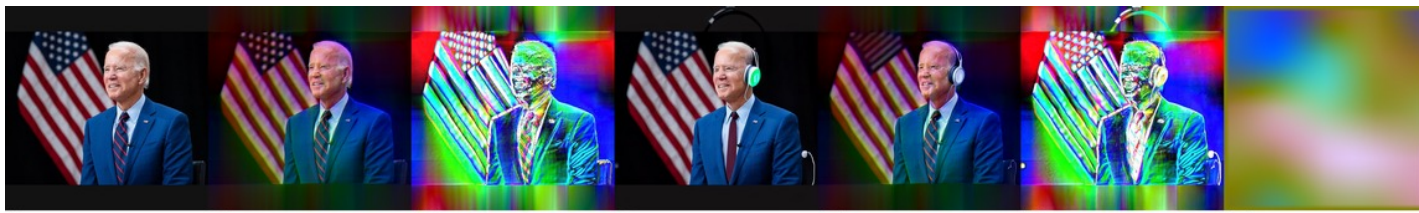
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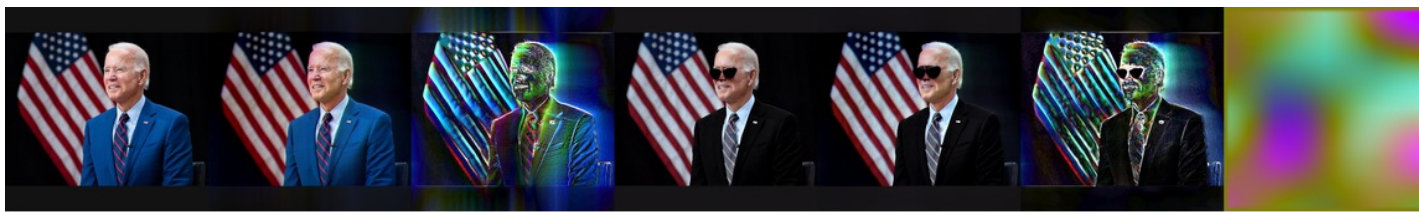
UNET.conv_in / add black sunglasses / joe_biden_01 - invalid_input_destroyed



VAE_image_latent / add headphones / joe_biden_01 - invalid_fft_saturated



VAE_image_latent / add black sunglasses / joe_biden_01 - valid_or_visually_check



VAE_image_latent / add black sunglasses / barack_obama_01 - valid_or_visually_check



VAE_image_latent / add headphones / barack_obama_01 - valid_or_visually_check



unet.conv_in / add black sunglasses / michelle_obama_01 - valid_or_visually_check



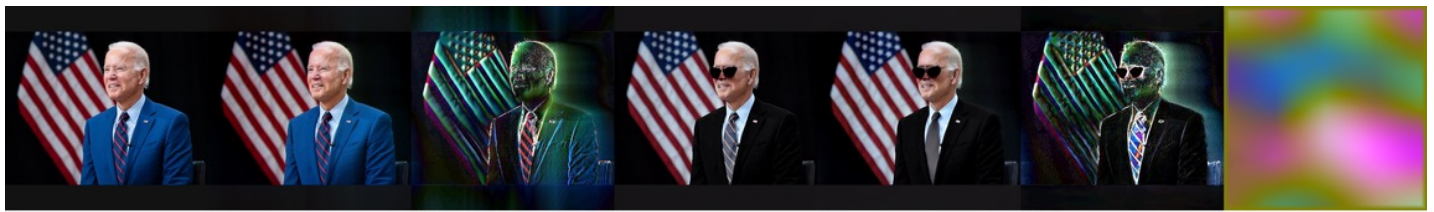
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unet.mid_block / add black sunglasses / barack_obama_01 - valid_or_visually_check



unet.mid_block / add black sunglasses / joe_biden_01 - valid_or_visually_check



unet.mid_block / add black sunglasses / kamala_harris_01 - valid_or_visually_check



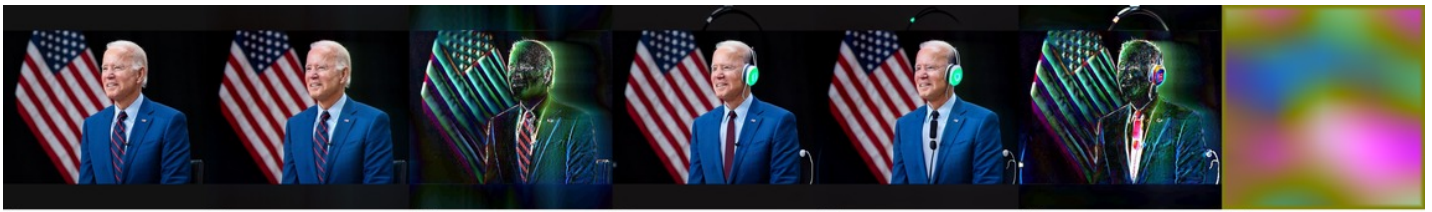
unet.mid_block / add black sunglasses / michelle_obama_01 - valid_or_visually_check



unet.mid_block / add headphones / barack_obama_01 - valid_or_visually_check



unet.mid_block / add headphones / joe_biden_01 - valid_or_visually_check



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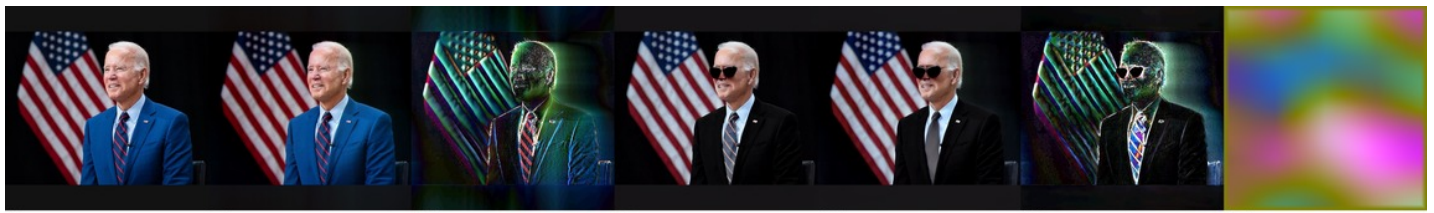
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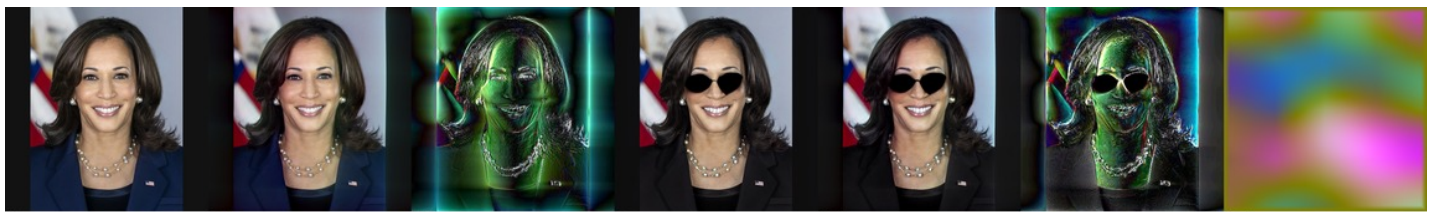
unet.up_blocks.0 / add black sunglasses / barack_obama_01 - valid_or_visually_check



unet.up_blocks.0 / add black sunglasses / joe_biden_01 - valid_or_visually_check



unet.up_blocks.0 / add black sunglasses / kamala_harris_01 - valid_or_visually_check



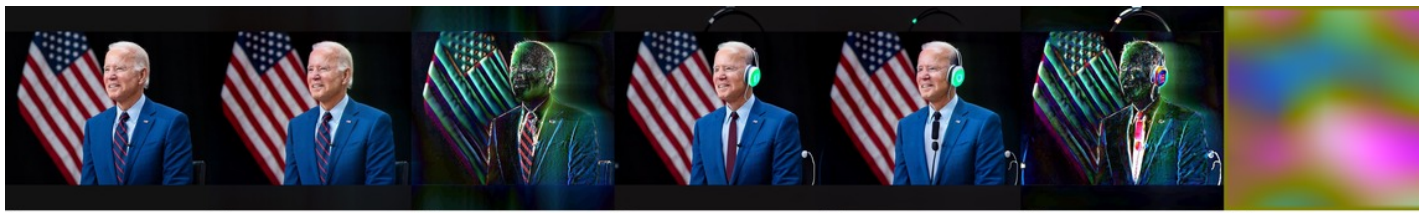
unet.up_blocks.0 / add black sunglasses / michelle_obama_01 - valid_or_visually_check



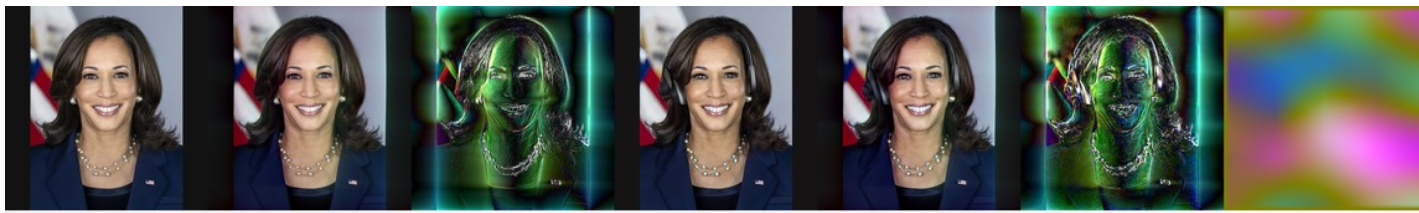
unet.up_blocks.0 / add headphones / barack_obama_01 - valid_or_visually_check



UNET.up_blocks.0 / add headphones / joe_biden_01 - valid_or_visually_check



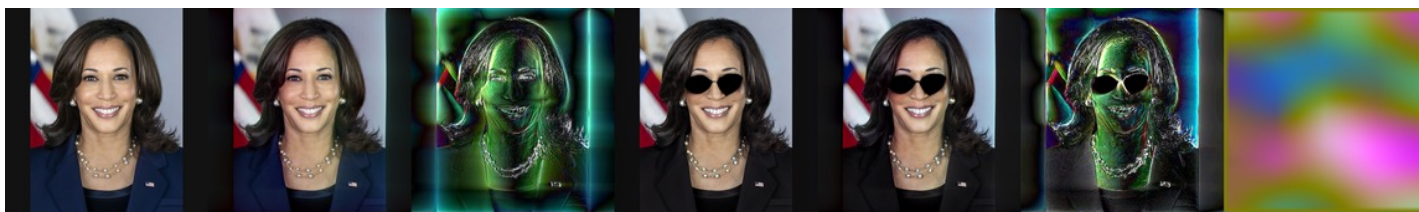
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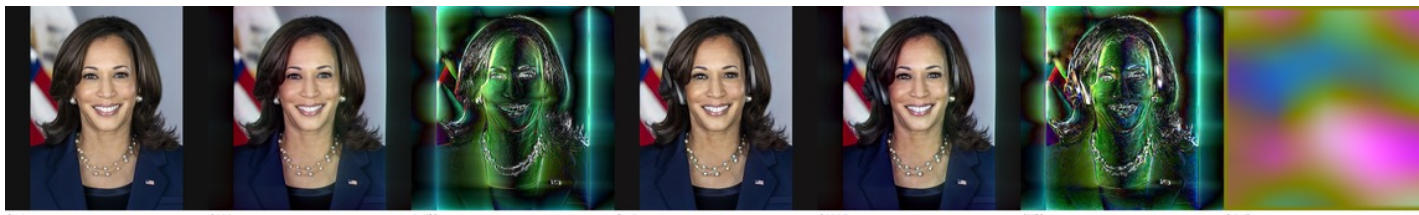
VAE_image_latent / add black sunglasses / kamala_harris_01 - valid_or_visually_check



VAE_image_latent / add black sunglasses / michelle_obama_01 - valid_or_visually_check



vae_image_latent / add headphones / kamala_harris_01 - valid_or_visually_check



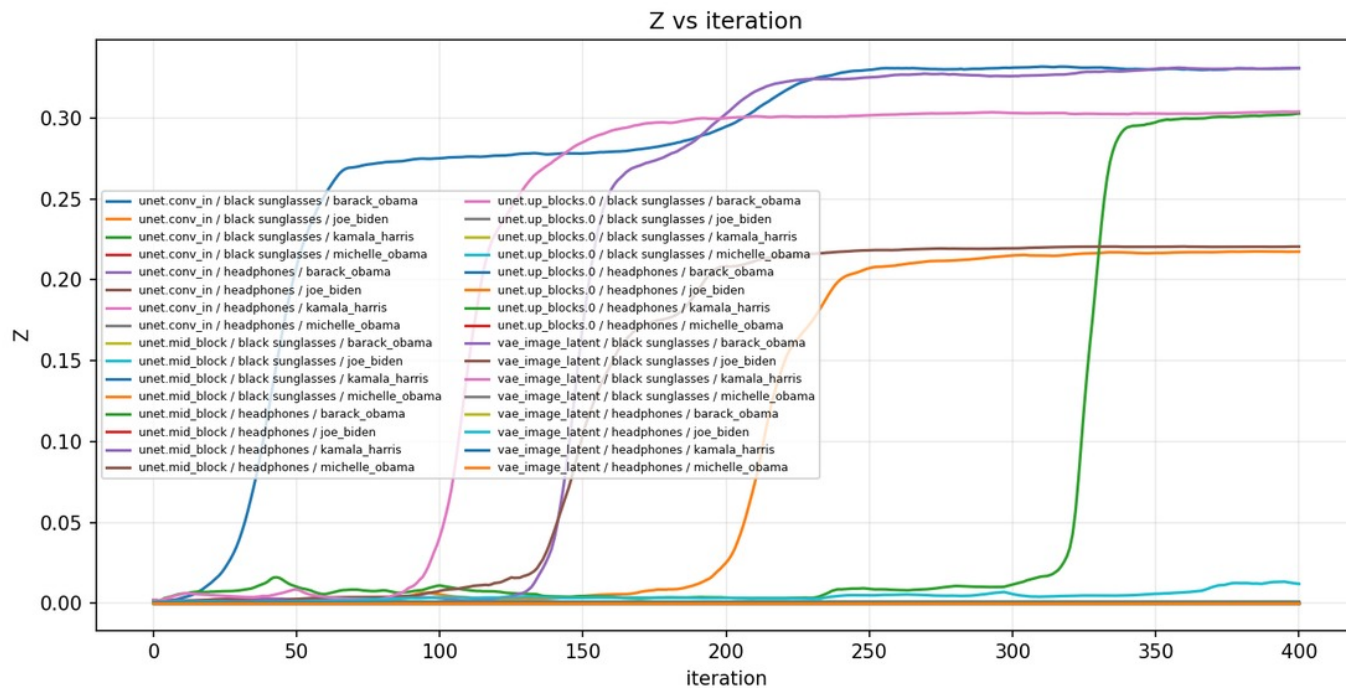
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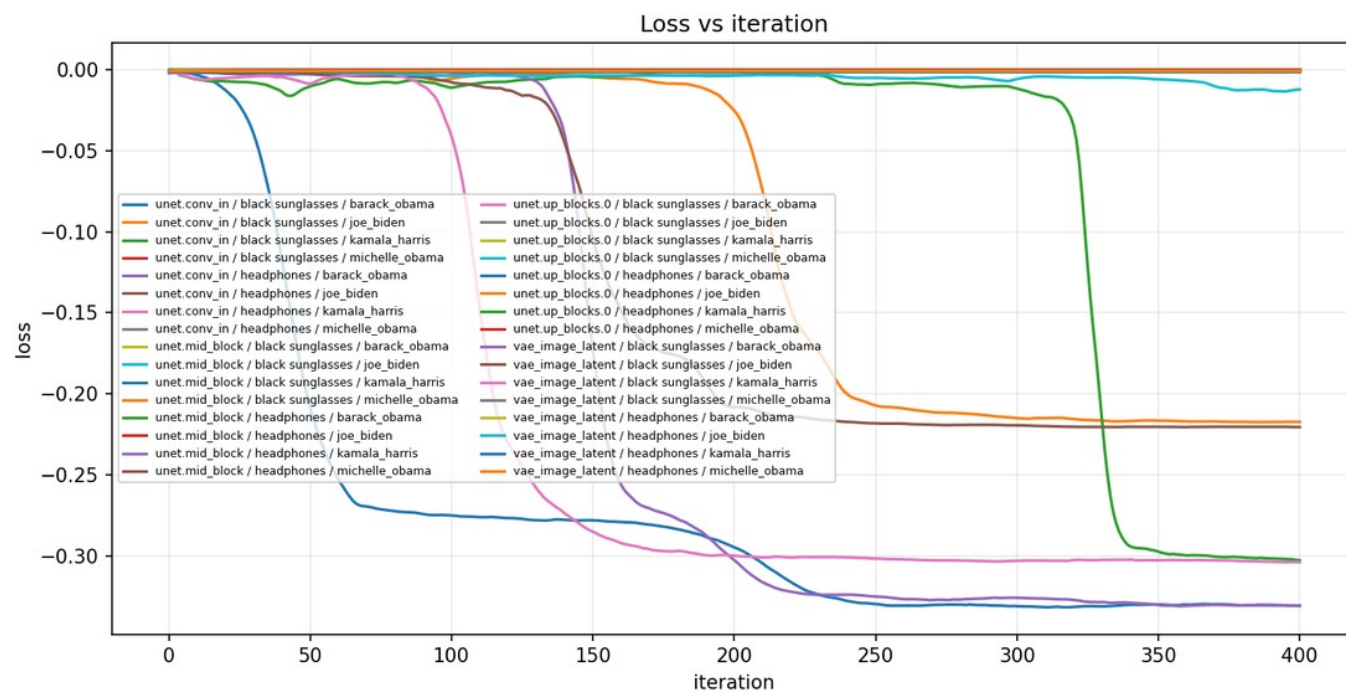
Graphs

Cross-run graphs

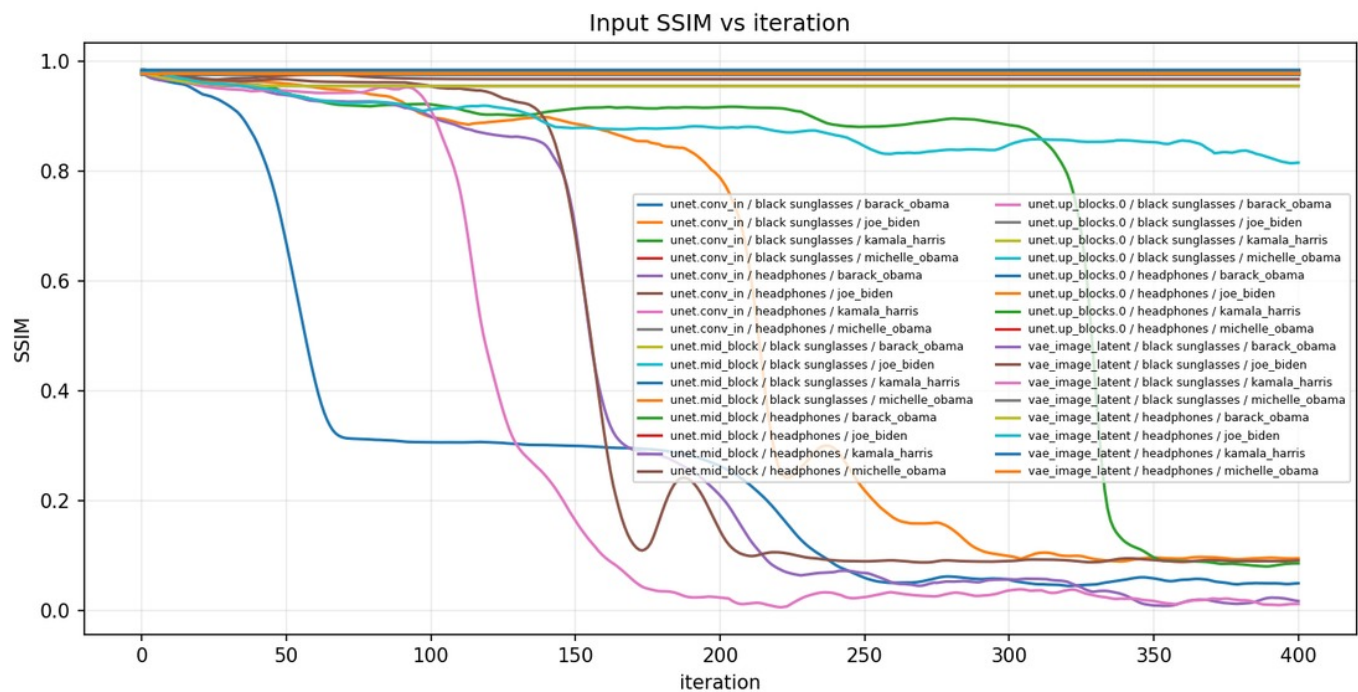
Z vs iteration



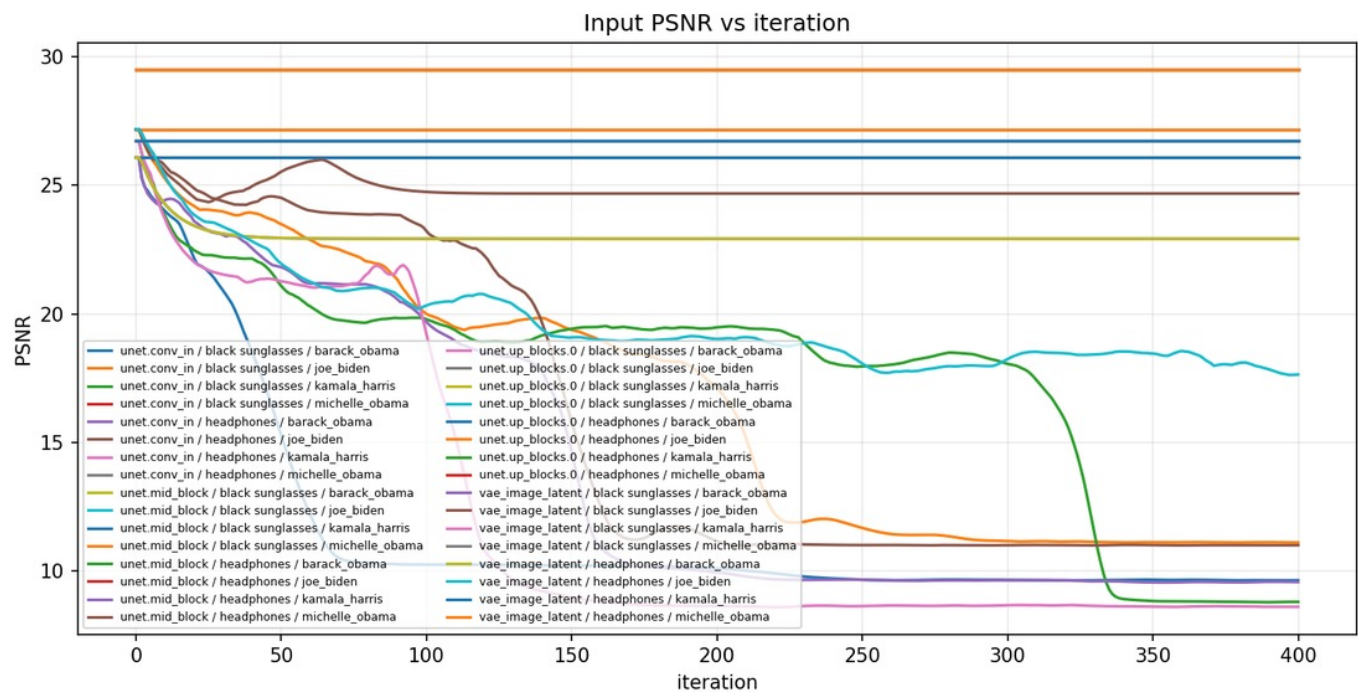
Loss vs iteration



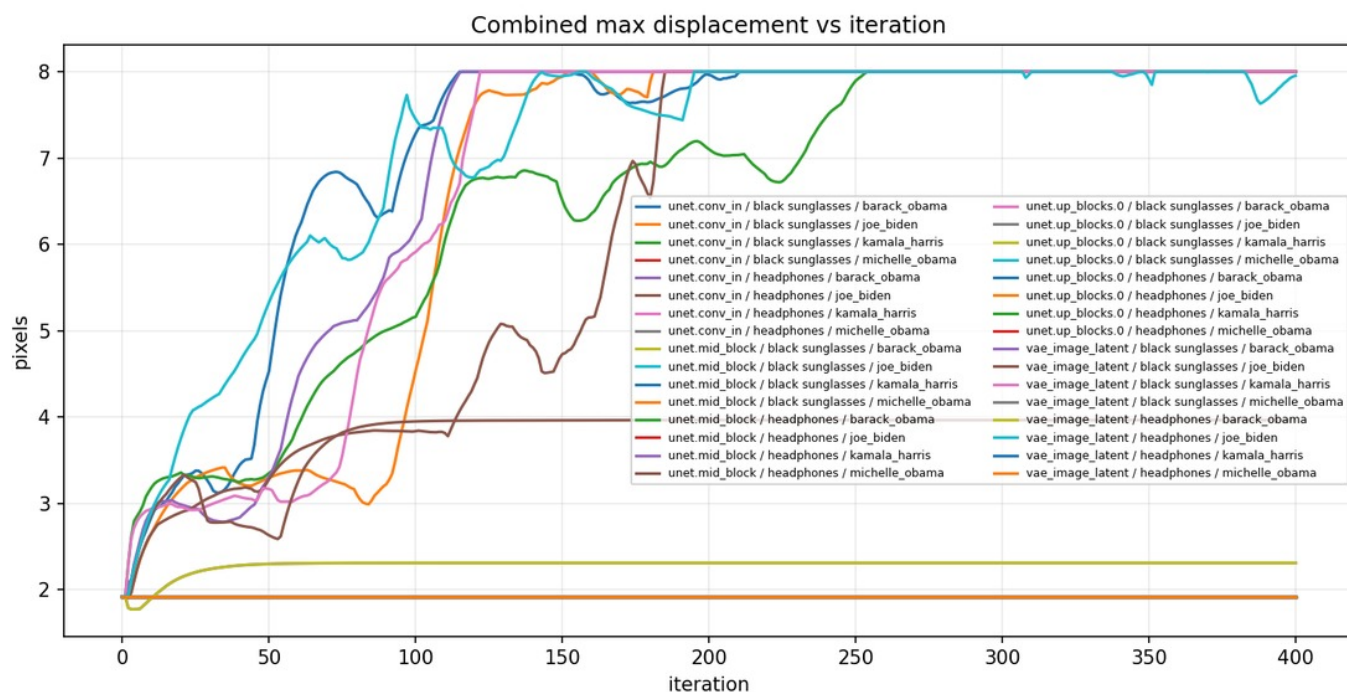
Input SSIM vs iteration



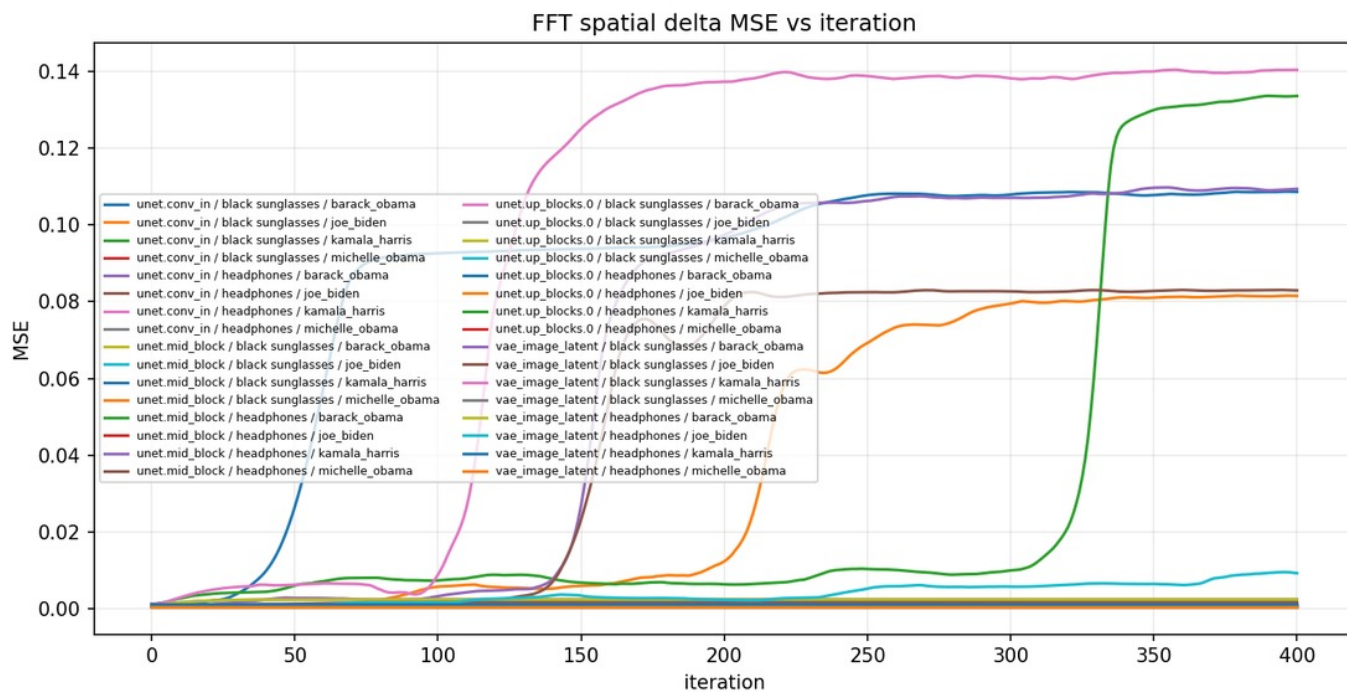
Input PSNR vs iteration



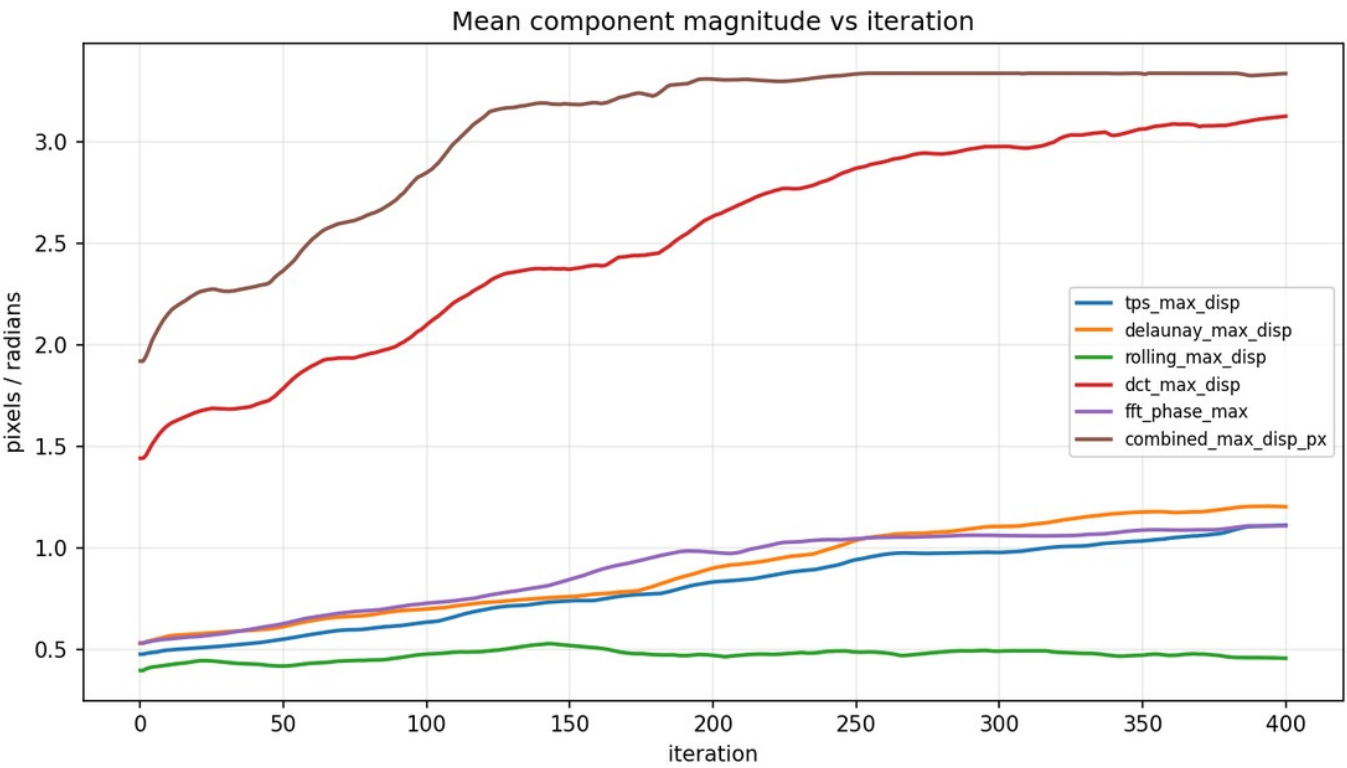
Combined max displacement vs iteration



FFT spatial delta MSE vs iteration



Component magnitude vs iteration



Z increase vs input SSIM

